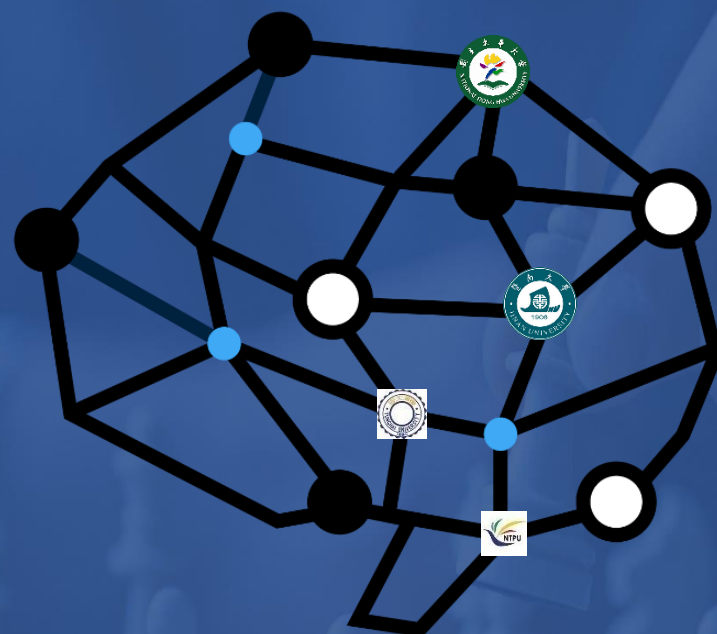
教育部課程改進計畫
人工智慧技術與應用 - 最先進之電腦對局遊戲系統設計



Chap. 3-2 Artificial Intelligence
Constraint Satisfaction problems

課程：人工智慧導論

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Outline

- What is CSP?
- Backtracking for CSP
- Local search for CSPs
- Problem structure and decomposition

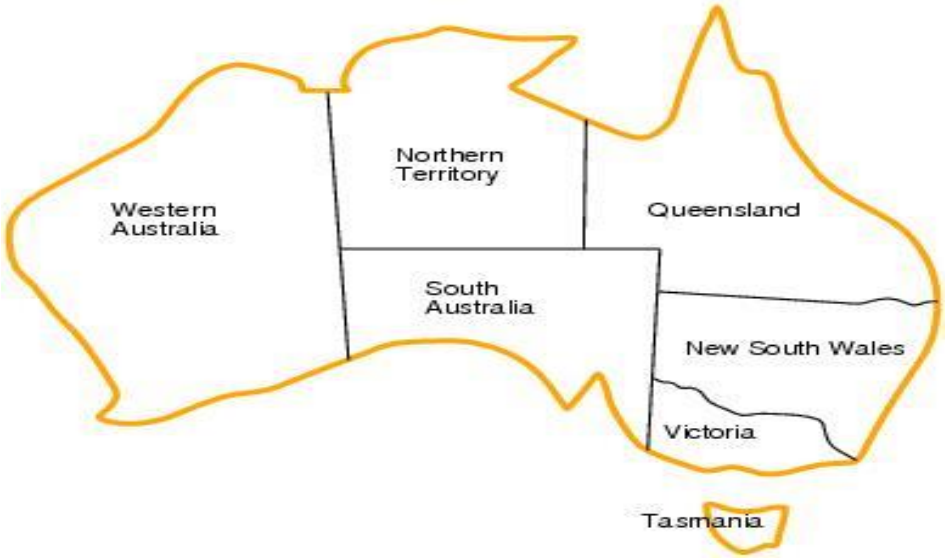
Constraint satisfaction problems

- What is a CSP?
  - **Finite set of variables $V_1, V_2, \ldots, V_n$**
  - **Finite set of constraints $C_1, C_2, \ldots, C_m$**
  - **Nonemtpy domain of possible values for each variable $D_{V1}, D_{V2}, \ldots D_{Vn}$**
  - **Each constraint C_i limits the values that variables can take, e.g., $V_1 \neq V_2$**
- A *state* is defined as an *assignment* of values to some or all variables.
- *Consistent assignment*: assignment does not violate the constraints.

Constraint satisfaction problems

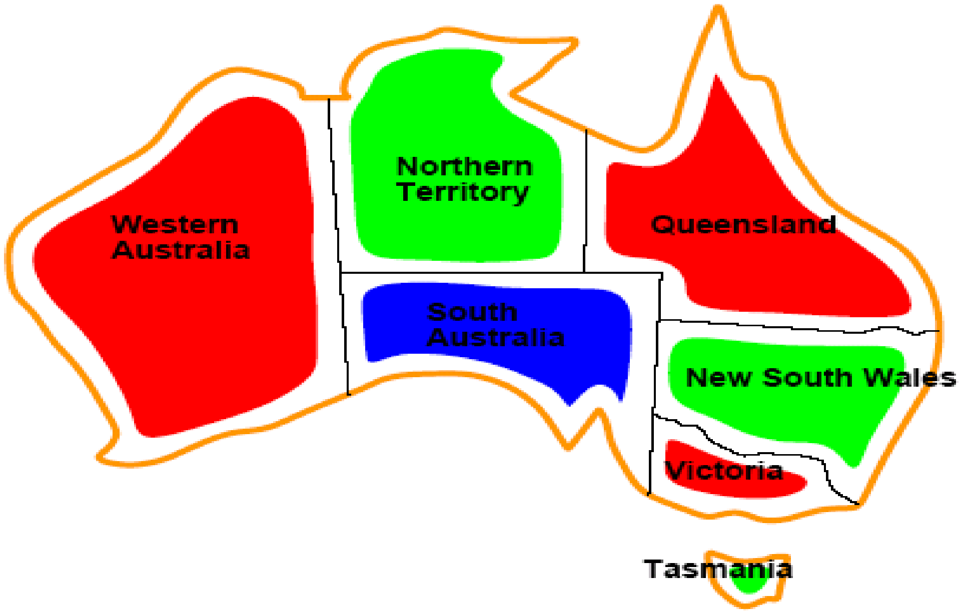
- An assignment is *complete* when every value is mentioned.
- A *solution* to a CSP is a complete assignment that satisfies all constraints.
- Some CSPs require a solution that maximizes an *objective function*.
- Applications: Scheduling the time of observations on the Hubble Space Telescope, Floor planning, Map coloring, Cryptography

CSP example: map coloring



- Variables: *WA, NT, Q, NSW, V, SA, T*
- Domains: $D_i=\{red,green,blue\}$
- Constraints:adjacent regions must have different colors.
  - E.g. $WA \neq NT$ (if the language allows this)
  - E.g. $(WA,NT) = \{(red,green),(red,blue),(green,red),\dots\}$

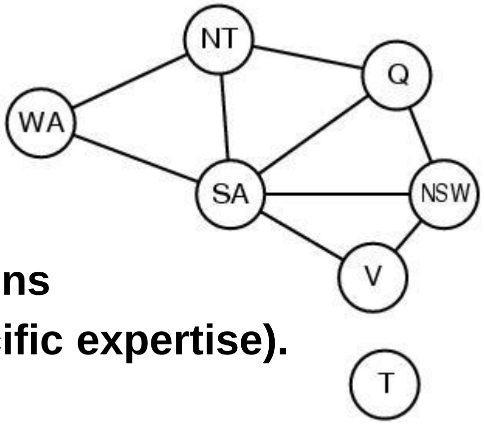
CSP example: map coloring



- Solutions are assignments satisfying all constraints, e.g.

  {WA=red,NT=green,Q=red,NSW=green,V=red,SA=blue,T=green}

Constraint graph



- CSP benefits
  - **Standard representation pattern**
  - **Generic goal and successor functions**
  - **Generic heuristics (no domain specific expertise).**

- Constraint graph = nodes are variables, edges show constraints.
  - **Graph can be used to simplify search.**
    - e.g. Tasmania is an independent subproblem.

Varieties of CSPs

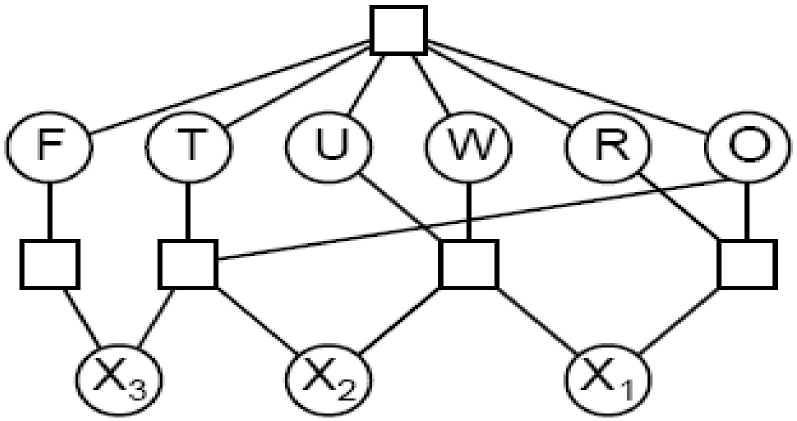
- Discrete variables
  - **Finite domains; size *d* ⇒$O(d^n)$ complete assignments.**
    - E.g. Boolean CSPs, include. Boolean satisfiability (NP-complete).
  - **Infinite domains (integers, strings, etc.)**
    - E.g. job scheduling, variables are start/end days for each job
    - Need a constraint language e.g $StartJob_1 + 5 \leq StartJob_3$.
    - Linear constraints solvable, nonlinear undecidable.
- Continuous variables
  - **e.g. start/end times for Hubble Telescope observations.**

Varieties of constraints

- Unary constraints involve a single variable.
  - **e.g. *SA* $\neq$ *green***
- Binary constraints involve pairs of variables.
  - **e.g. *SA* $\neq$ *WA***
- Higher-order constraints involve 3 or more variables.
  - **e.g. cryptharithmetic column constraints.**
- Preference (soft constraints) e.g. *red* is better than *green* often representable by a cost for each variable assignment $\rightarrow$ constrained optimization problems.

Example: cryptharithmetic

$$\begin{array}{cccc} & T & W & O \\ + & T & W & O \\ \hline F & O & U & R \end{array}$$



Variables: $F\ T\ U\ W\ R\ O\ X_1\ X_2\ X_3$
Domains: $\{0,1,2,3,4,5,6,7,8,9\}$
Constraints

alldiff(F,T,U,W,R,O)
$O+O=R+10\cdot X_1$, etc.

CSP as a standard search problem

- A CSP can easily expressed as a standard search problem.
- Incremental formulation
  - ***Initial State*: the empty assignment {}.**
  - ***Successor function*: Assign value to unassigned variable provided that there is not conflict.**
  - ***Goal test*: the current assignment is complete.**
  - ***Path cost*: as constant cost for every step.**

CSP as a standard search problem

- This is the same for all CSP's !!!
- Solution is found at depth n (if there are n variables).
  - **Hence depth first search can be used.**
- Path is irrelevant, so complete state representation can also be used.
- Branching factor b at the top level is nd.
- $b=(n-l)d$ at depth l, hence $n!d^n$ leaves (only d^n complete assignments).

Commutativity

- CSPs are commutative.
  - **The order of any given set of actions has no effect on the outcome.**
  - **Example: choose colors for Australian territories one at a time**
    - [WA=red then NT=green] same as [NT=green then WA=red]
    - All CSP search algorithms consider a single variable assignment at a time $\Rightarrow$ there are d^n leaves.

Backtracking search

- Cfr. Depth-first search
- Chooses values for one variable at a time and backtracks when a variable has no legal values left to assign.
- Uninformed algorithm
  - **No good general performance (see table p. 143, #3)**
- Add informed backtracking

Backtracking search

function BACKTRACKING-SEARCH(*csp*) **return** a solution or failure

return RECURSIVE-BACKTRACKING(*{}* , *csp*)

function RECURSIVE-BACKTRACKING(*assignment, csp*) **return** a solution or failure

if *assignment* is complete **then return** *assignment*

var ← SELECT-UNASSIGNED-VARIABLE(VARIABLES[*csp*],*assignment*,*csp*)

for each *value* **in** ORDER-DOMAIN-VALUES(*var, assignment, csp*) **do**

if *value* is consistent with *assignment* according to CONSTRAINTS[*csp*] **then**

add *{var=value}* to assignment

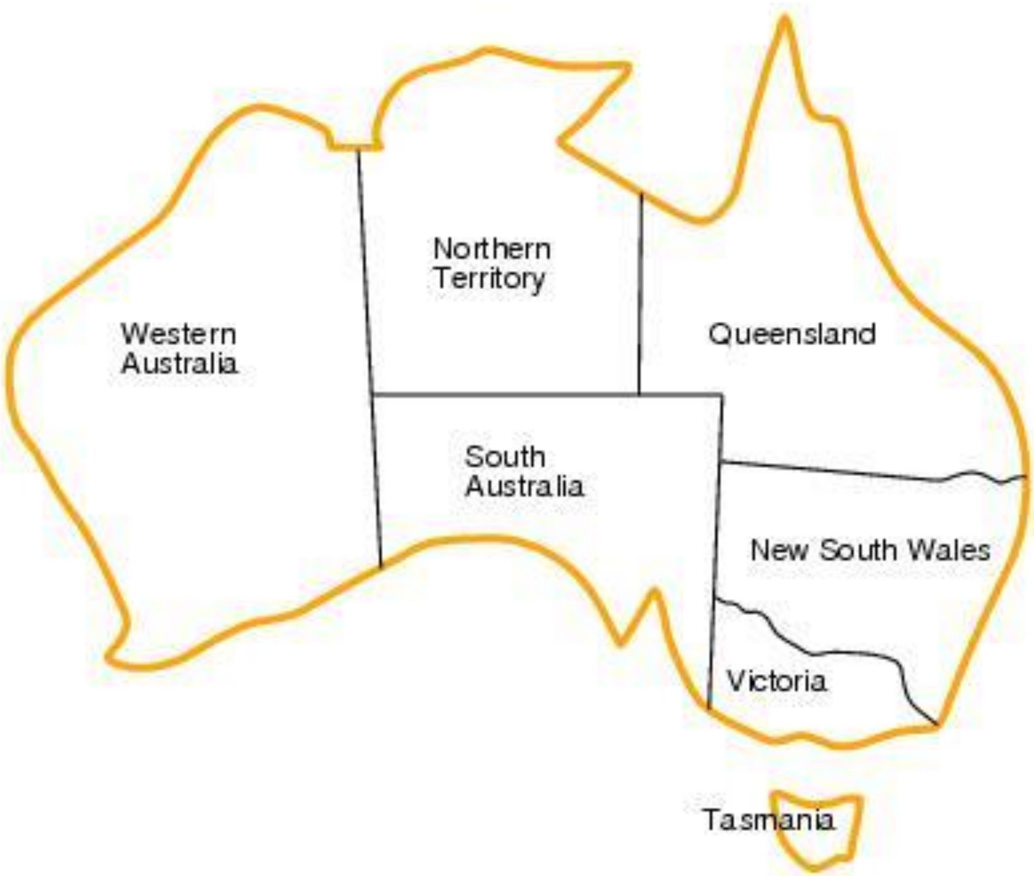
result ← RRECURSIVE-BACTRACKING(*assignment, csp*)

if *result* ≠ *failure* **then return** *result*

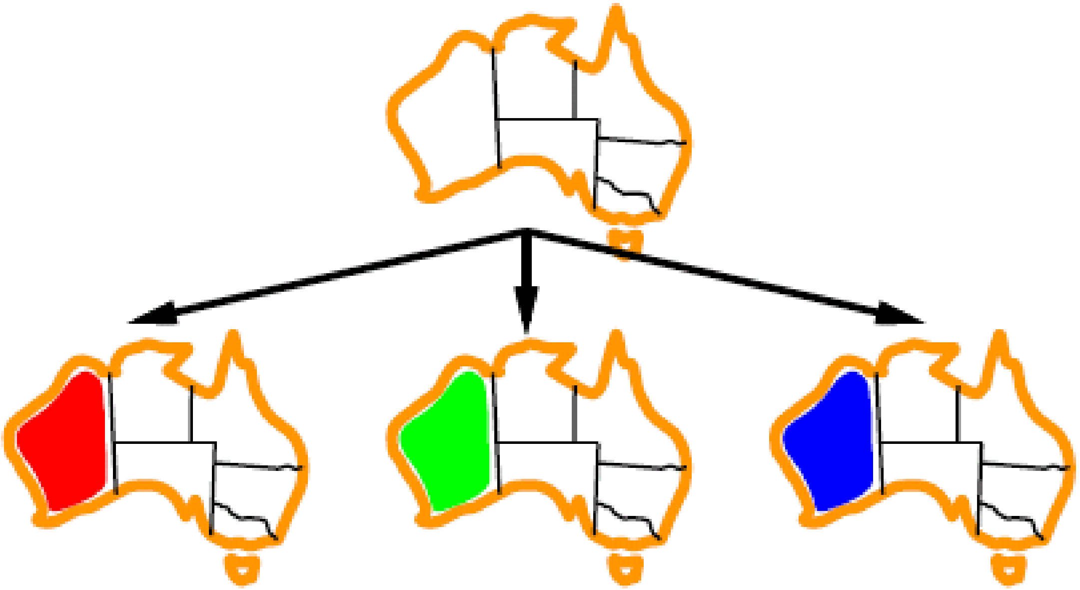
remove *{var=value}* from *assignment*

return *failure*

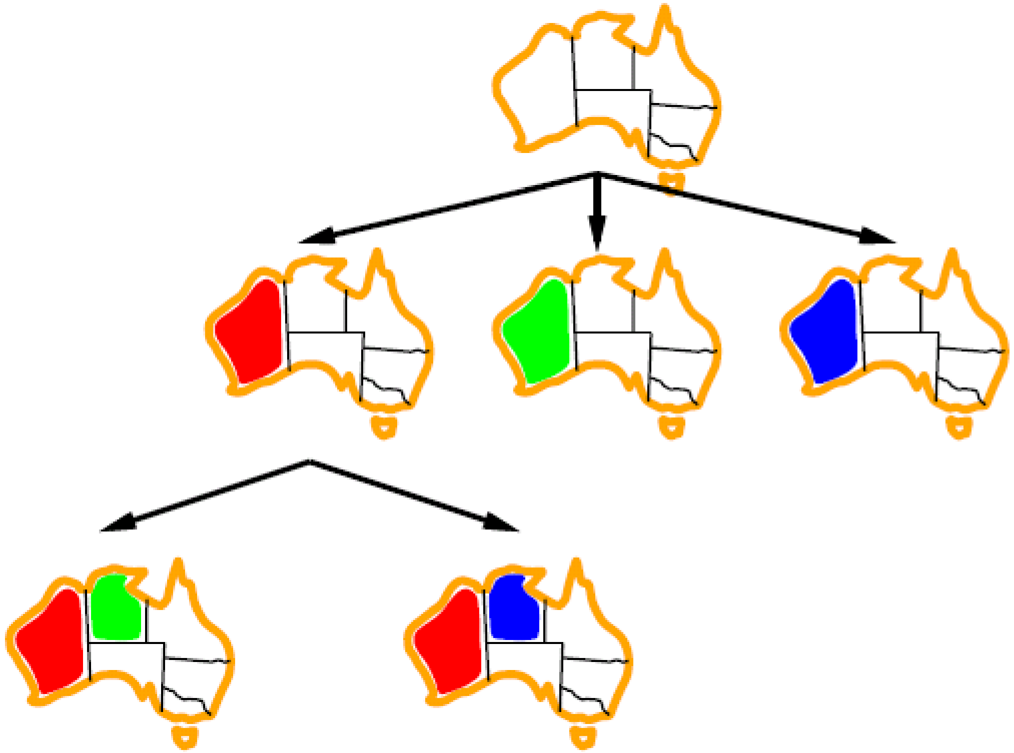
Backtracking example



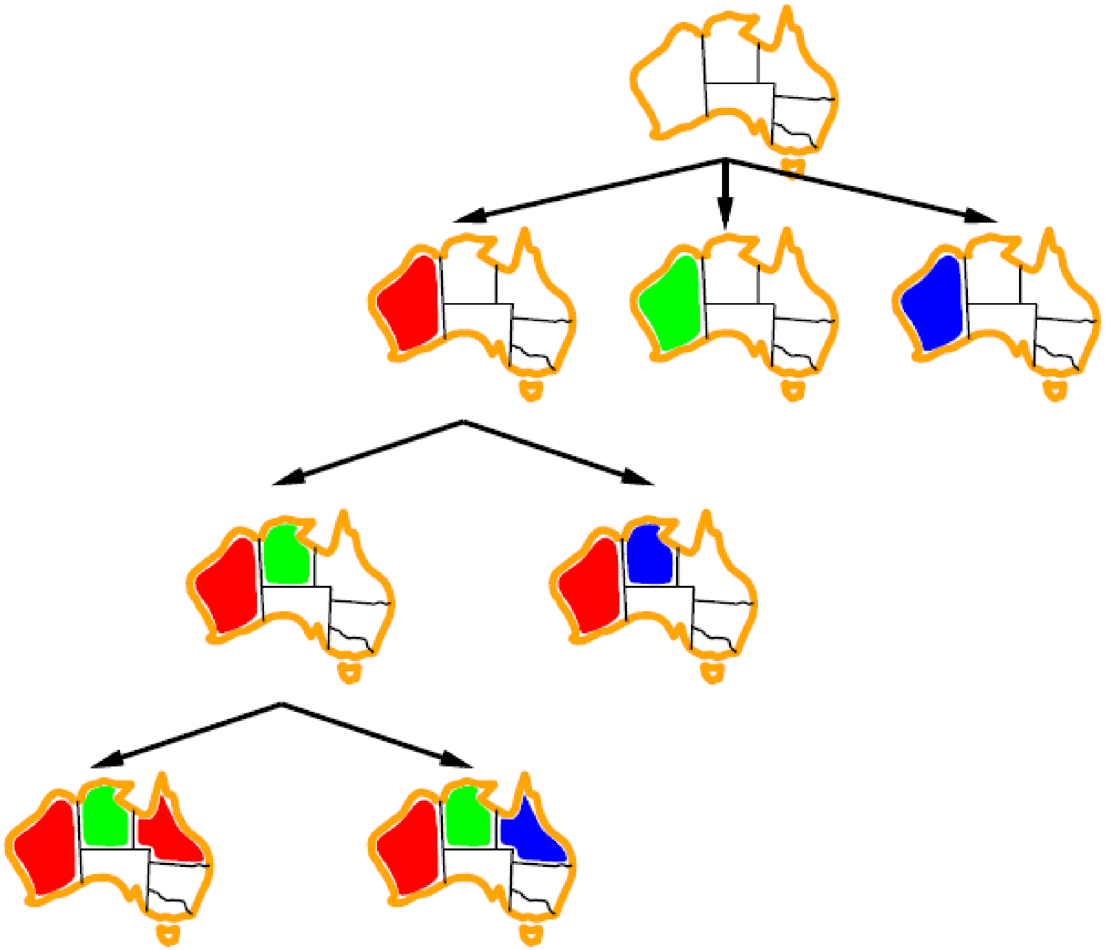
Backtracking example



Backtracking example



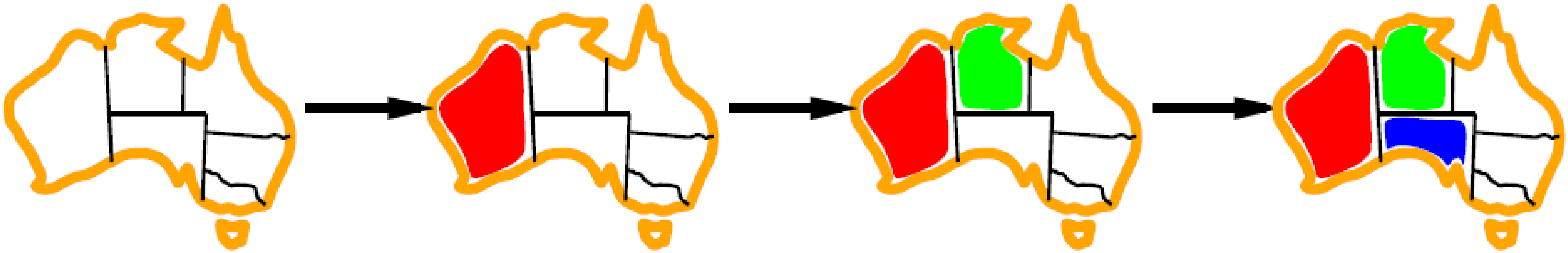
Backtracking example



Improving backtracking efficiency

- Previous improvements $\rightarrow$ introduce heuristics
- General-purpose methods can give huge gains in speed:
  - **Which variable should be assigned next?**
  - **In what order should its values be tried?**
  - **Can we detect inevitable failure early?**
  - **Can we take advantage of problem structure?**

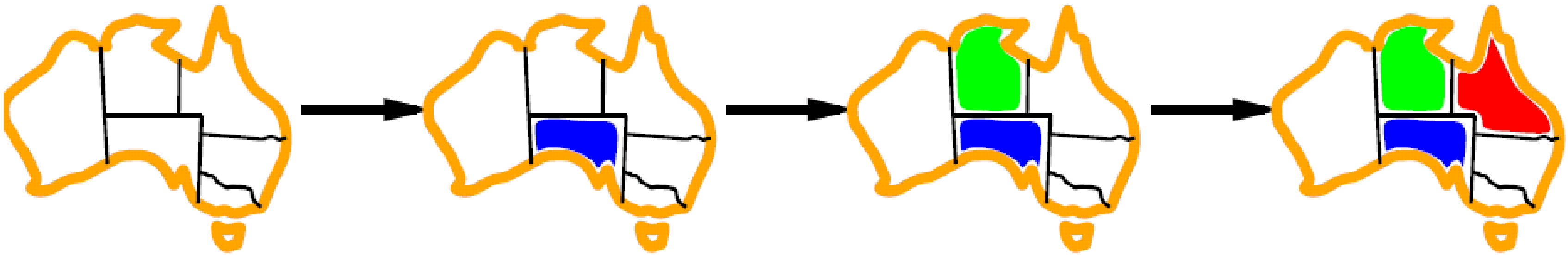
Minimum remaining values



var ← SELECT-UNASSIGNED-VARIABLE(VARIABLES[*csp*],*assignment*,*csp*)

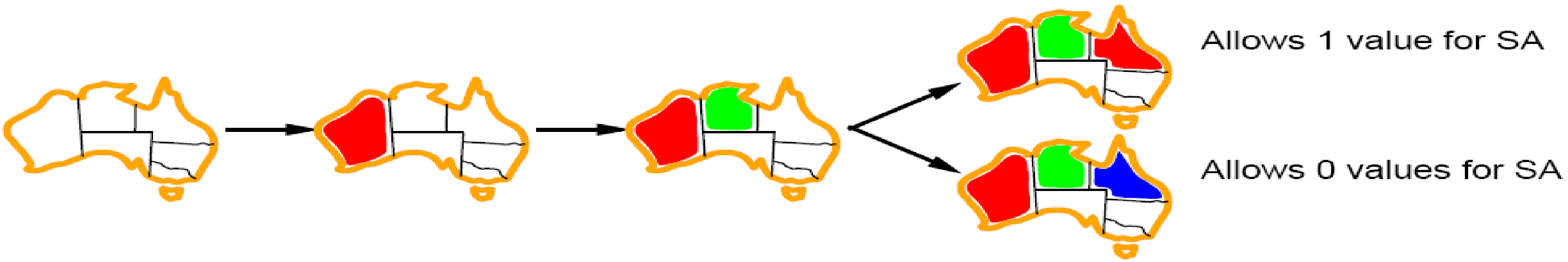
- A.k.a. most constrained variable heuristic
- *Rule*: choose variable with the fewest legal moves
- *Which variable shall we try first? SA!*

Degree heuristic



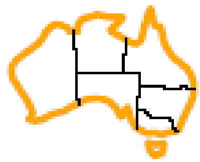
- Use degree heuristic
- *Rule*: select variable that is involved in the largest number of constraints on other unassigned variables.
- Degree heuristic is very useful as a tie breaker.
- *In what order should its values be tried?*

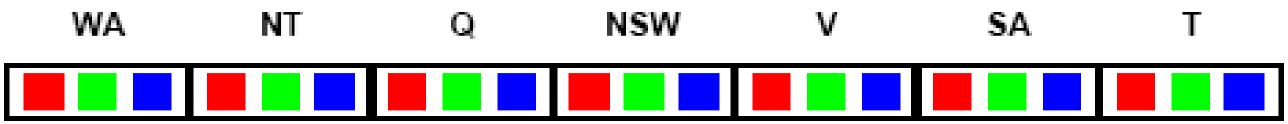
Least constraining value



- Least constraining value heuristic
- Rule: given a variable choose the least constraing value i.e. the one that leaves the maximum flexibility for subsequent variable assignments.

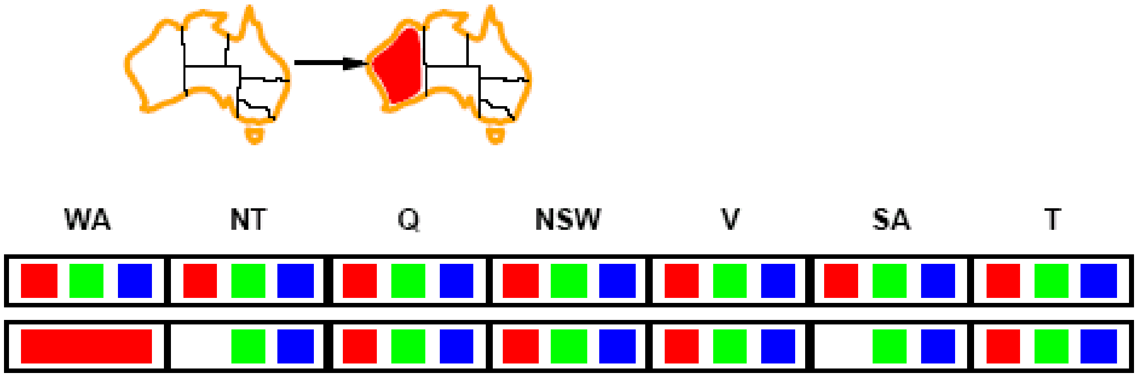
Forward checking





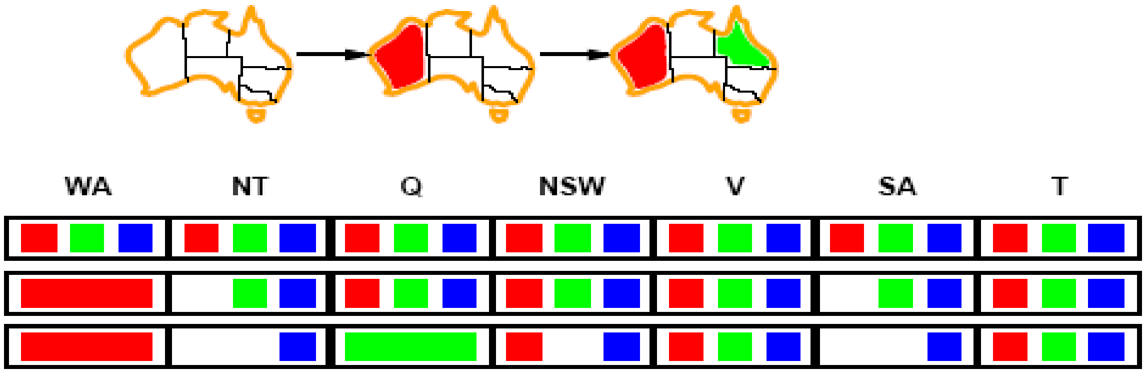
- Can we detect inevitable failure early?
  - *And avoid it later?*
- *Forward checking idea:* keep track of remaining legal values for unassigned variables.
- Terminate search when any variable has no legal values.

Forward checking



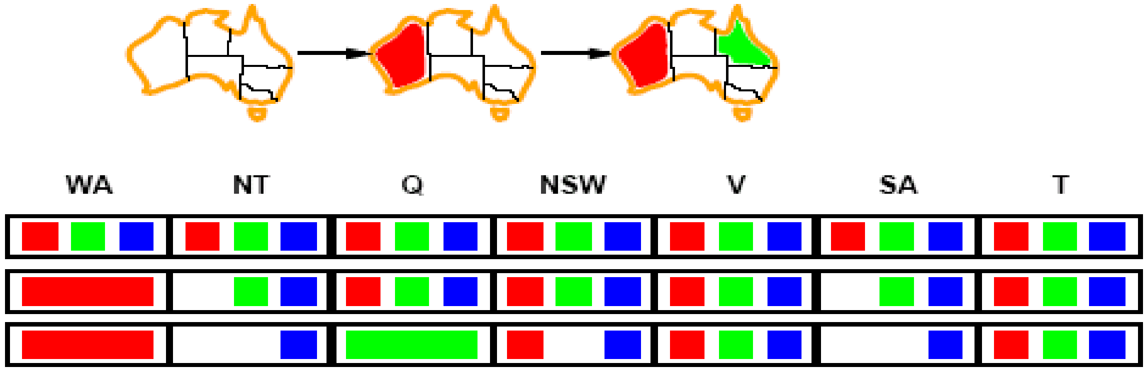
- Assign *{WA=red}*
- Effects on other variables connected by constraints with WA
  - ***NT can no longer be red***
  - ***SA can no longer be red***

Forward checking



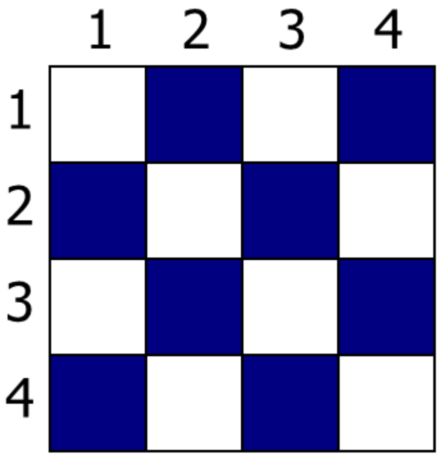
- Assign *{Q=green}*
- Effects on other variables connected by constraints with WA
    - ***NT can no longer be green***
    - ***NSW can no longer be green***
    - ***SA can no longer be green***

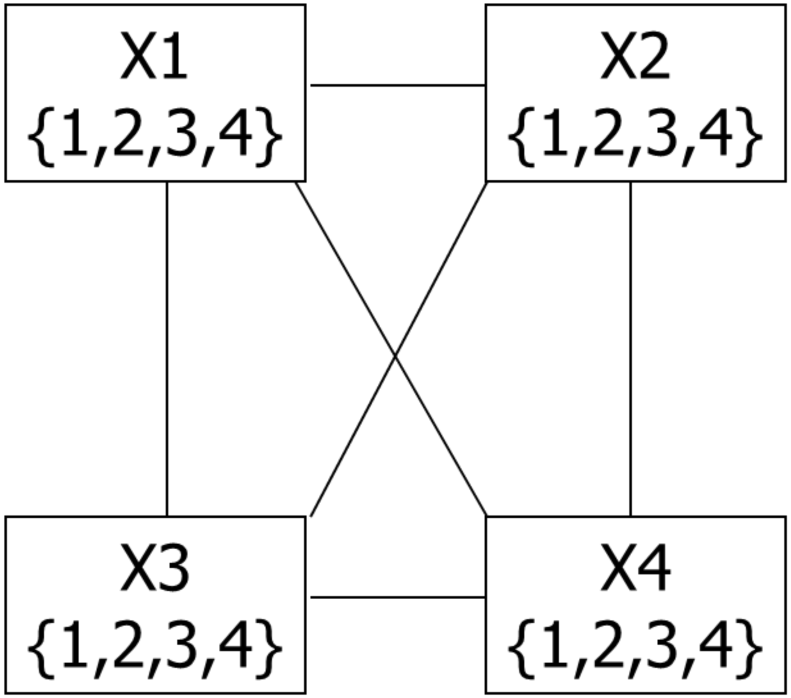
Forward checking



- If *V* is assigned *blue*
- Effects on other variables connected by constraints with WA
    - ***SA is empty***
    - ***NSW can no longer be blue***
- FC has detected that partial assignment is *inconsistent* with the constraints and backtracking can occur.

Example: 4-Queens Problem





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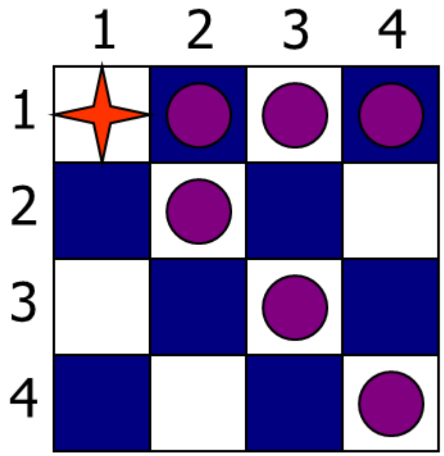

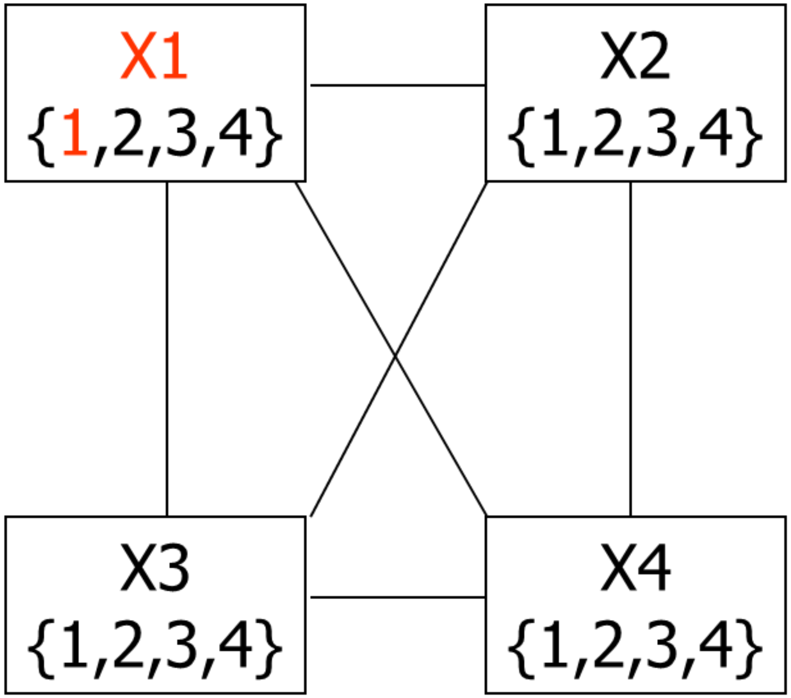

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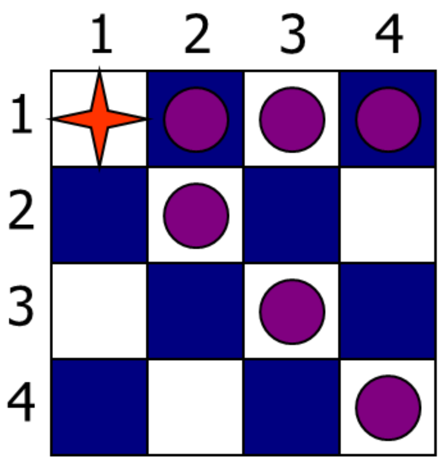

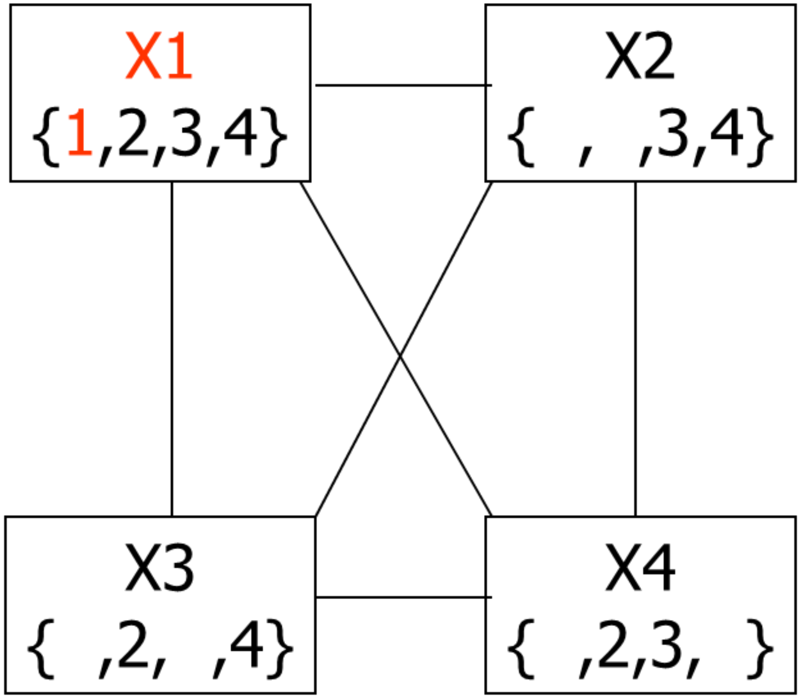

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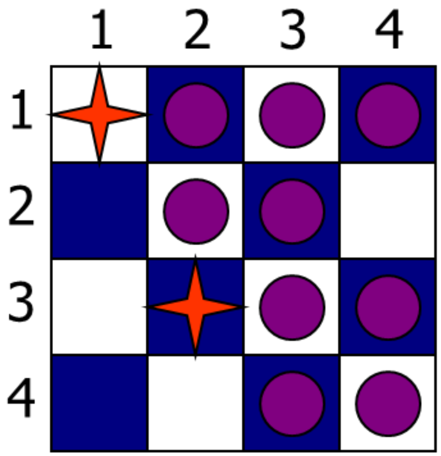

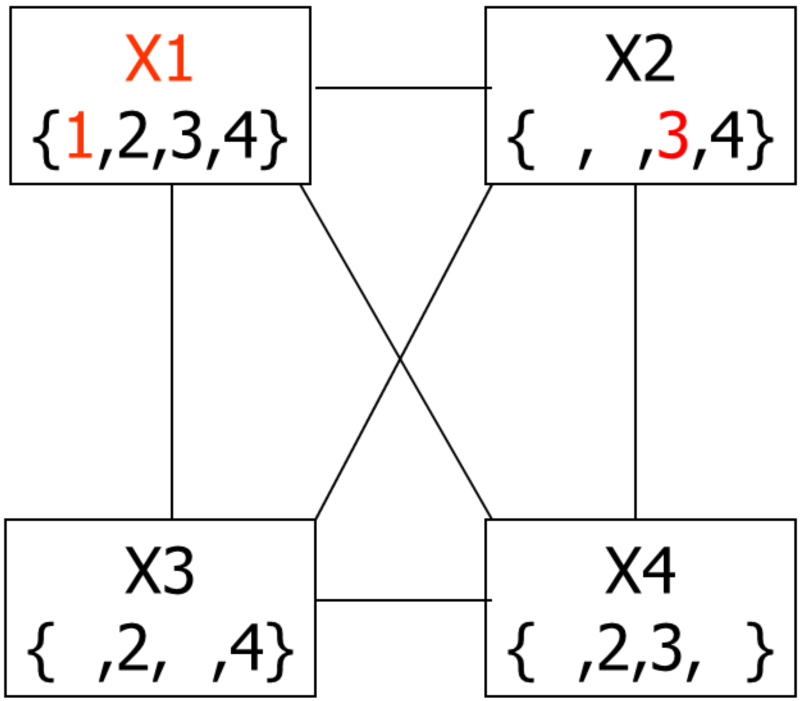

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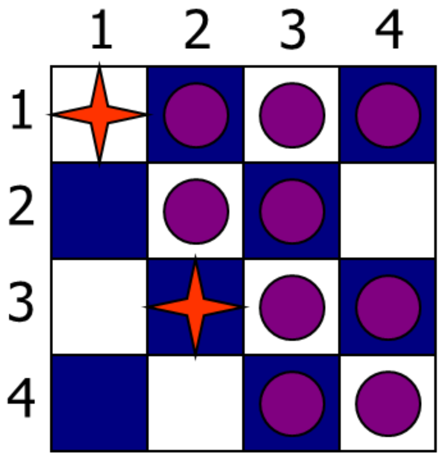

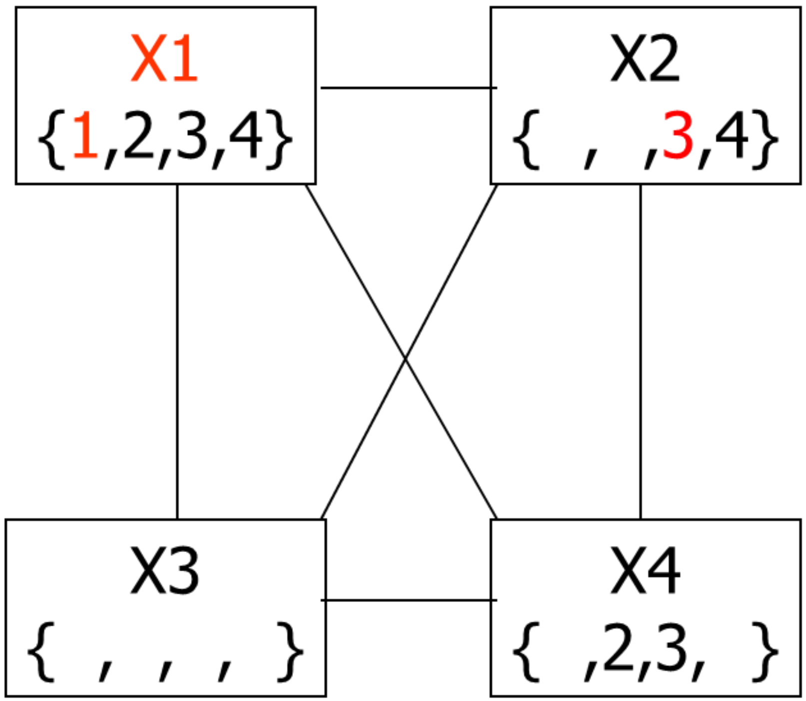

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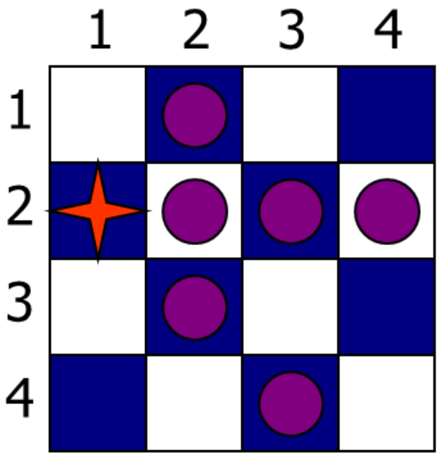

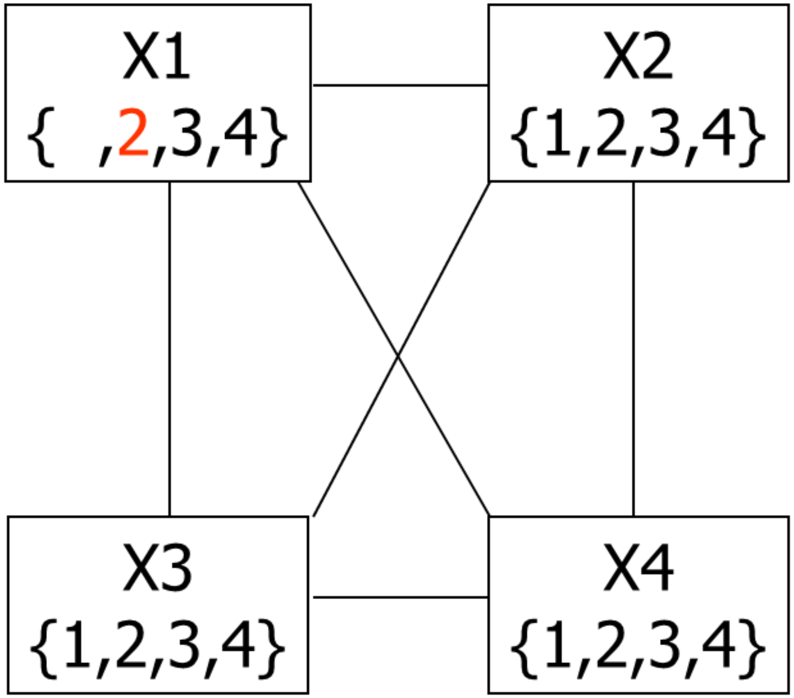

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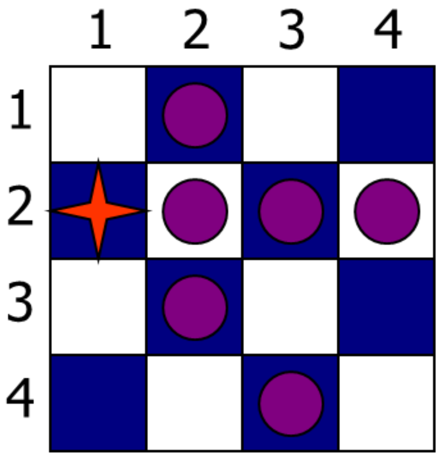

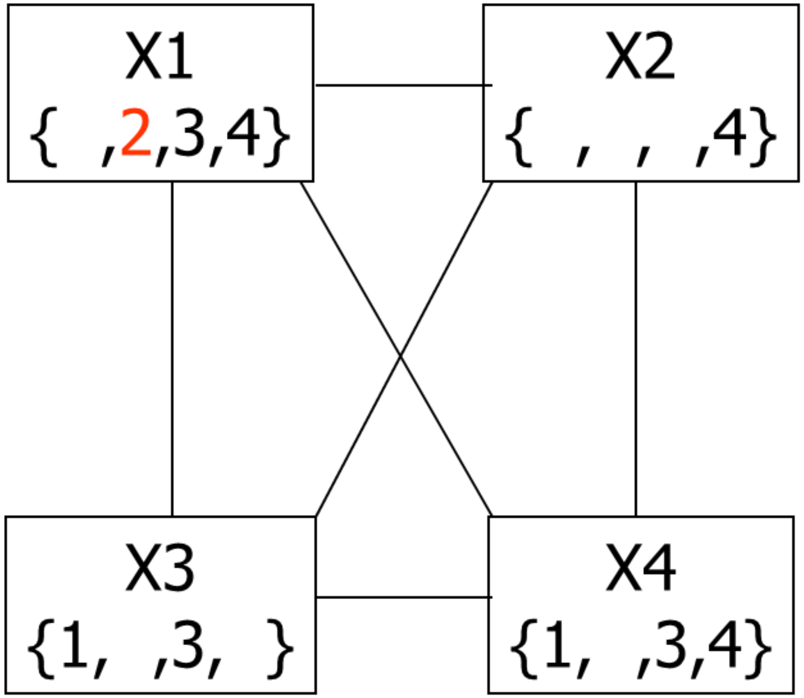

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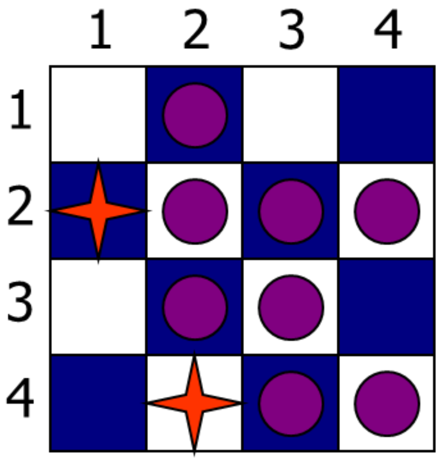

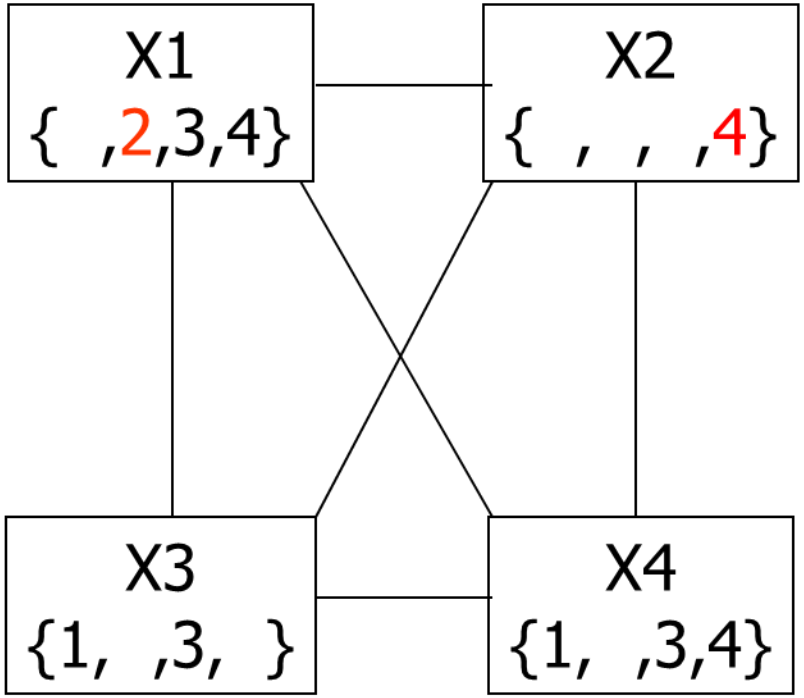

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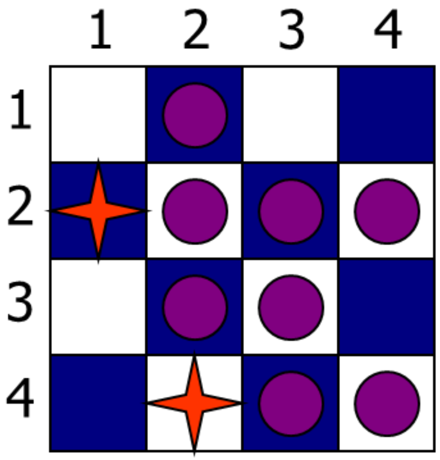

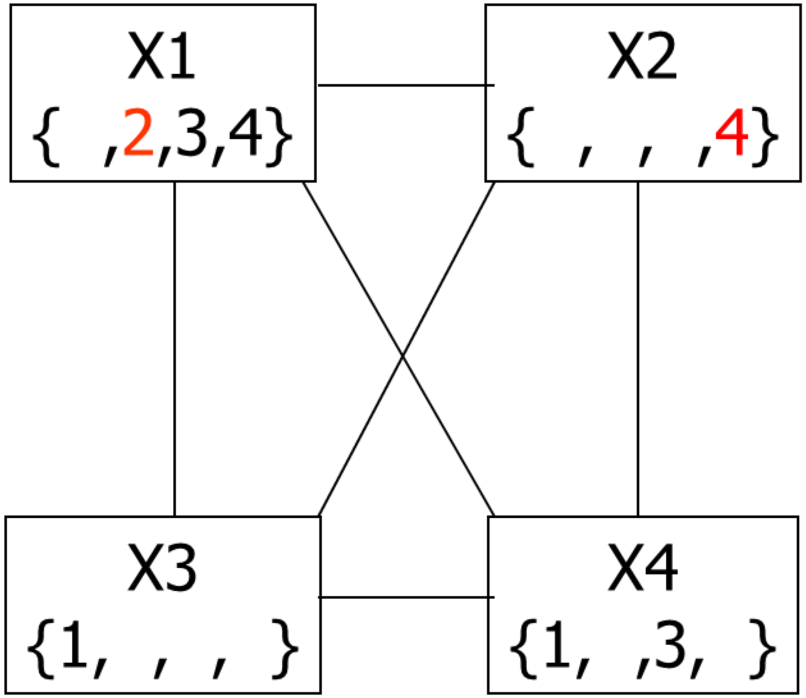

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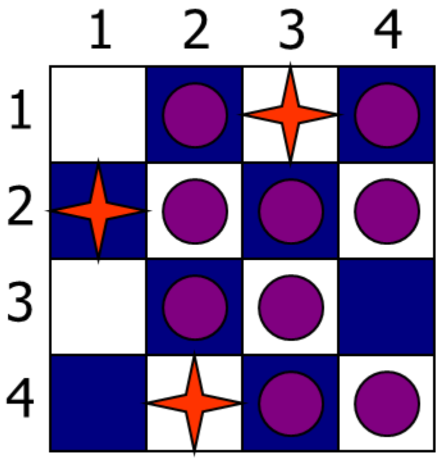

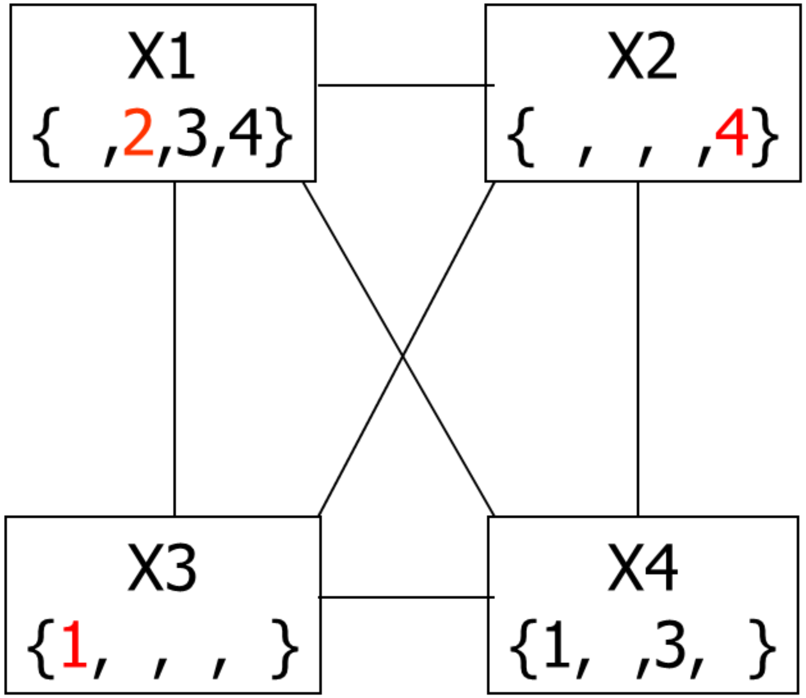

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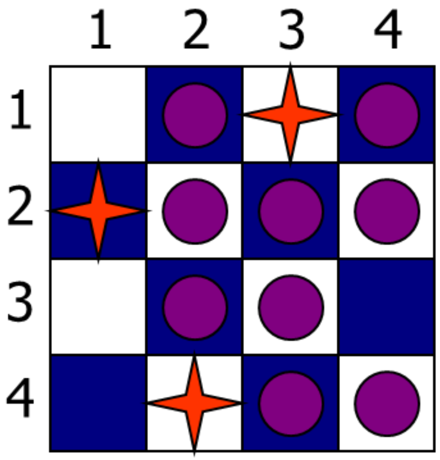

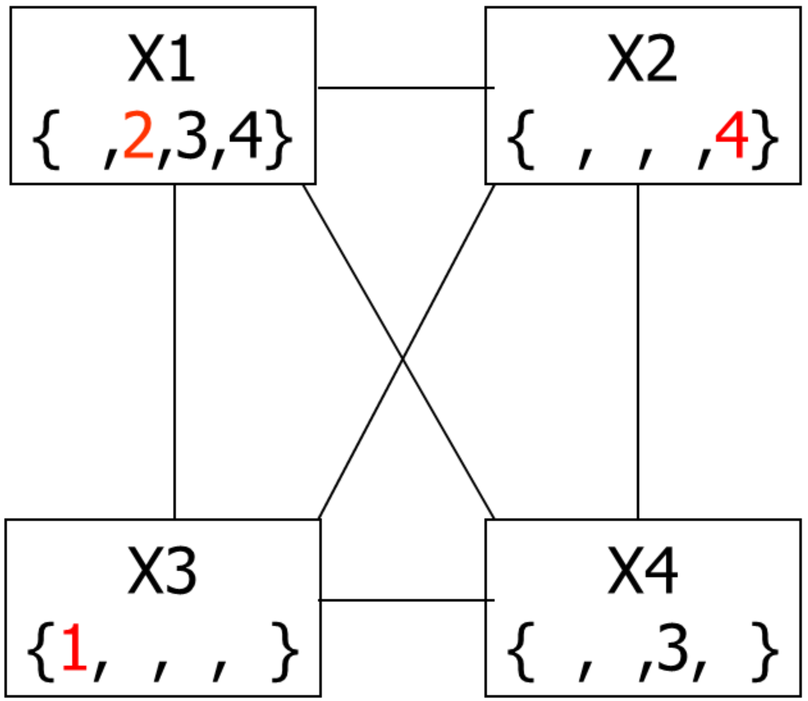

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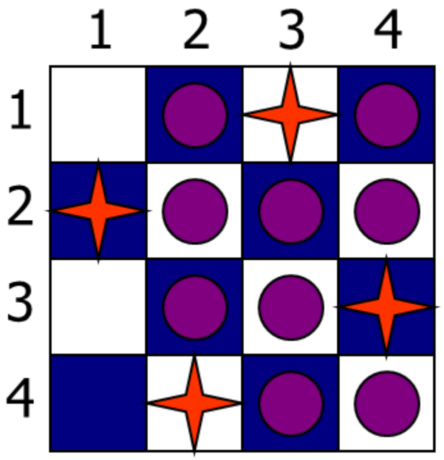
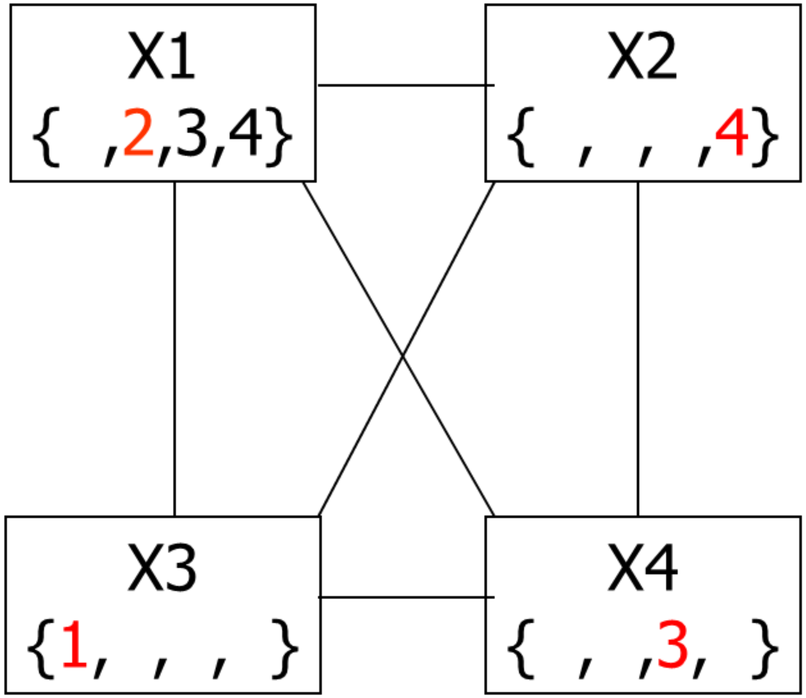
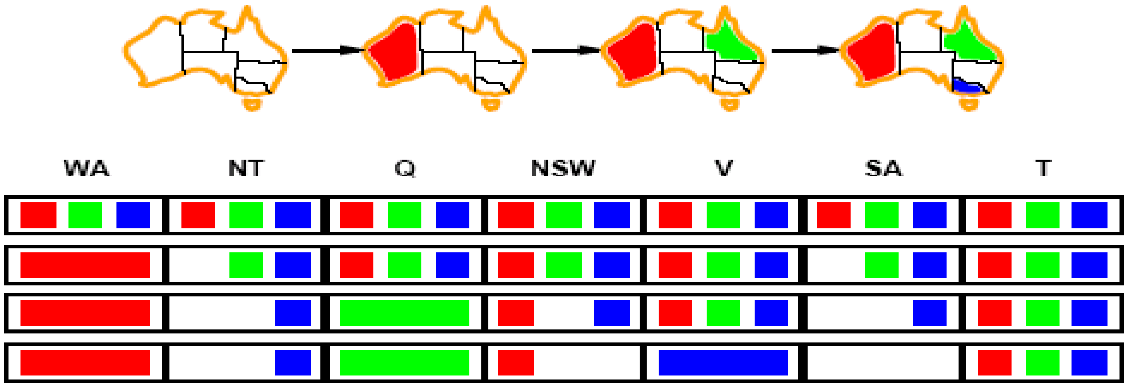


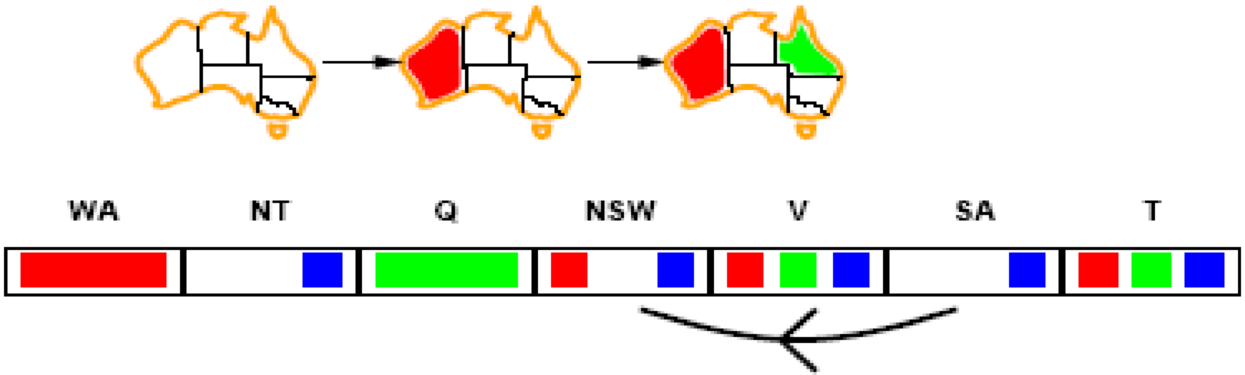


Constraint propagation



- Solving CSPs with combination of heuristics plus forward checking is more efficient than either approach alone.
- FC checking propagates information from assigned to unassigned variables but does not provide detection for all failures.
  - **NT and SA cannot be blue!**
- Constraint propagation repeatedly enforces constraints locally

Arc consistency



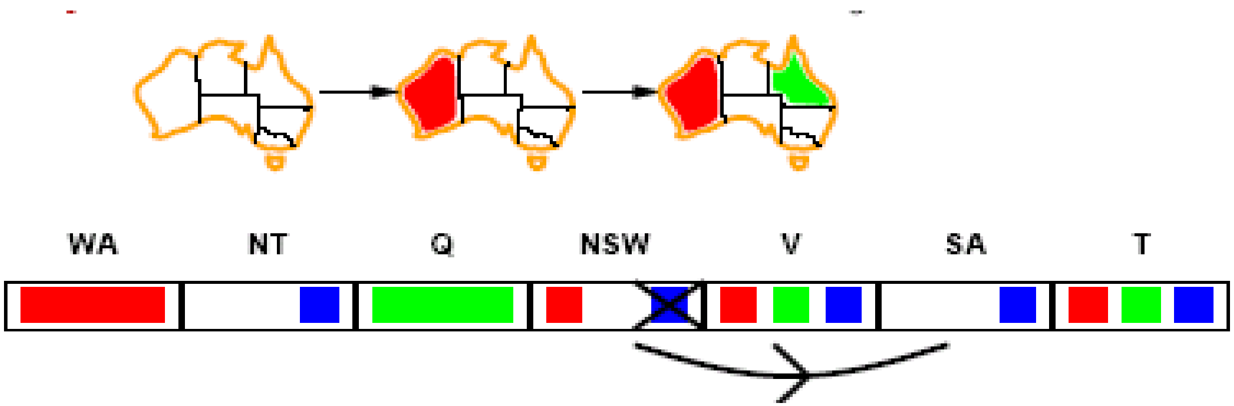
- $X \to Y$ is consistent iff

  for *every* value *x* of *X* there is some allowed *y*

- *SA* $\to$ *NSW* is consistent iff

  SA=blue and *NSW=red*

Arc consistency



- $X \rightarrow Y$ is consistent iff

  for *every* value *x* of *X* there is some allowed *y*

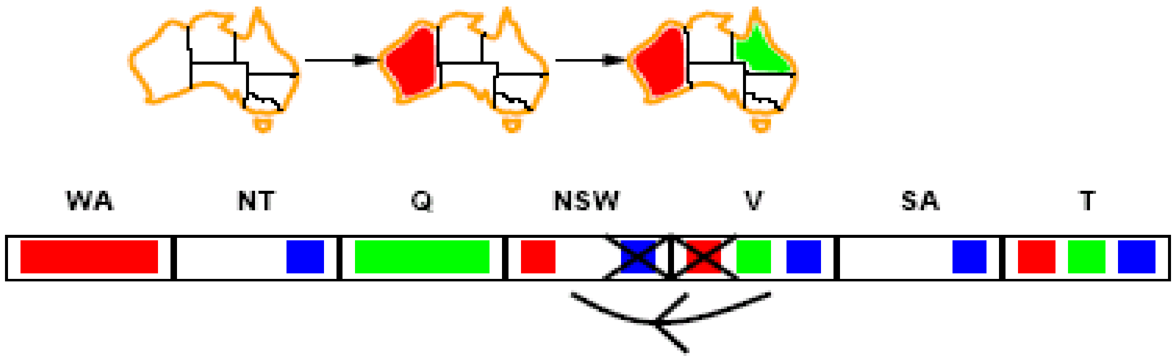
- *NSW* $\rightarrow$ *SA* is consistent iff

  NSW=red and *SA=blue*

  NSW=blue and SA=???

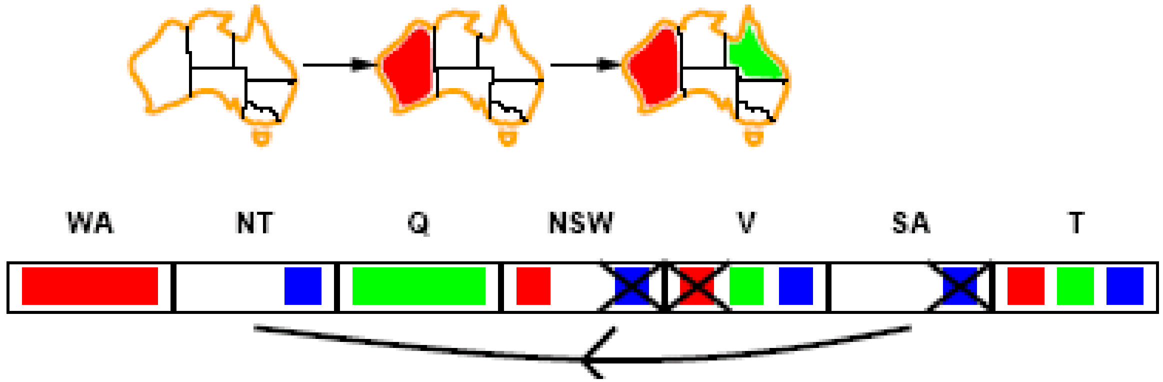
Arc can be made consistent by removing *blue* from *NSW*

Arc consistency



- Arc can be made consistent by removing *blue* from *NSW*
- RECHECK neighbours !!
    - **Remove red from *V***

Arc consistency



- Arc can be made consistent by removing *blue* from *NSW*
- RECHECK neighbours !!
  - **Remove red from *V***
- Arc consistency detects failure earlier than FC
- Can be run as a preprocessor or after each assignment.
  - **Repeated until no inconsistency remains**

Arc consistency algorithm

function AC-3(*csp*) **return** the CSP, possibly with reduced domains

inputs: *csp*, a binary csp with variables $\{X_1, X_2, \ldots, X_n\}$

local variables: *queue,* a queue of arcs initially the arcs in *csp*

while queue is not empty **do**

$(X_i, X_j) \leftarrow$ REMOVE-FIRST(*queue*)

if REMOVE-INCONSISTENT-VALUES(X_i, X_j) **then**

for each X_k **in** NEIGHBORS[X_i] **do**

add (X_k, X_i) to queue

function REMOVE-INCONSISTENT-VALUES(X_i, X_j) **return** *true* iff we remove a value

removed $\leftarrow$ *false*

for each *x* **in** DOMAIN[X_i] **do**

if no value *y* in DOMAIN[X_j] allows (x,y) to satisfy the constraints between X_i and X_j

then delete x from DOMAIN[X_i]; *removed* $\leftarrow$ *true*

return *removed*

Local search for CSP

- Use complete-state representation
- For CSPs
  - **allow states with unsatisfied constraints**
  - **operators reassign variable values**
- Variable selection: randomly select any conflicted variable
- Value selection: *min-conflicts heuristic*
  - **Select new value that results in a minimum number of conflicts with the other variables**

Local search for CSP

function MIN-CONFLICTS(*csp, max_steps*) **return** solution or failure

inputs: *csp*, a constraint satisfaction problem

max_steps, the number of steps allowed before giving up

current ← an initial complete assignment for *csp*

for *i* = 1 to *max_steps* **do**

if *current* is a solution for *csp* then return *current*

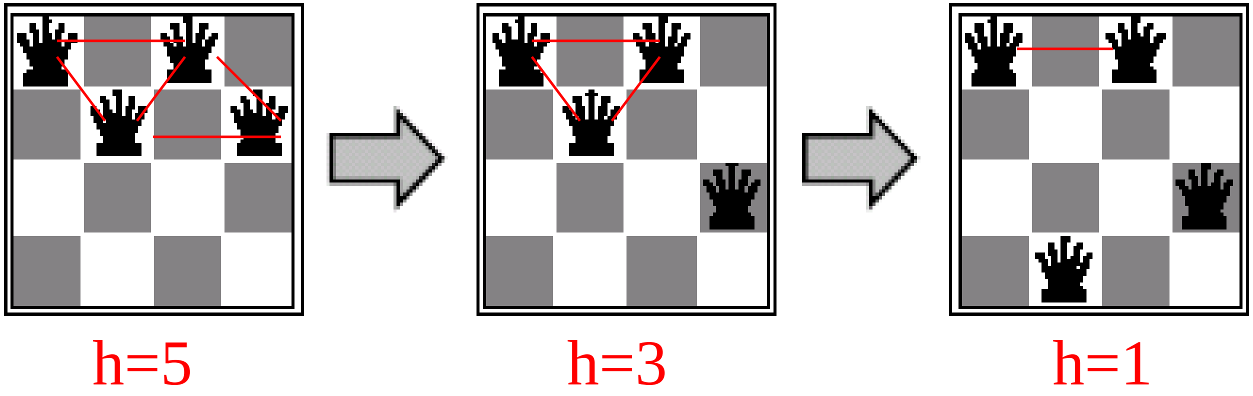
var ← a randomly chosen, conflicted variable from VARIABLES[*csp*]

value ← the value *v* for *var* that minimizes CONFLICTS(*var,v,current,csp*)

set *var* = *value* in *current*

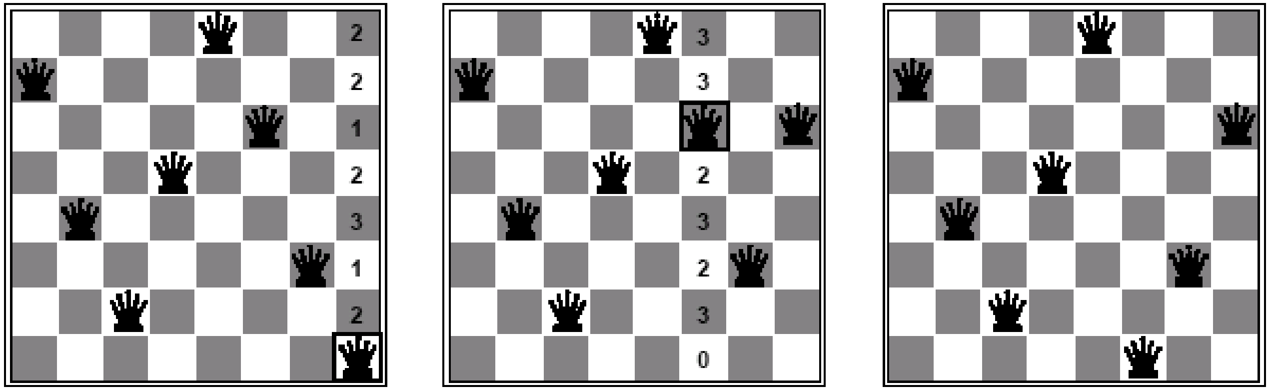
return *faiilure*

Min-conflicts example 1



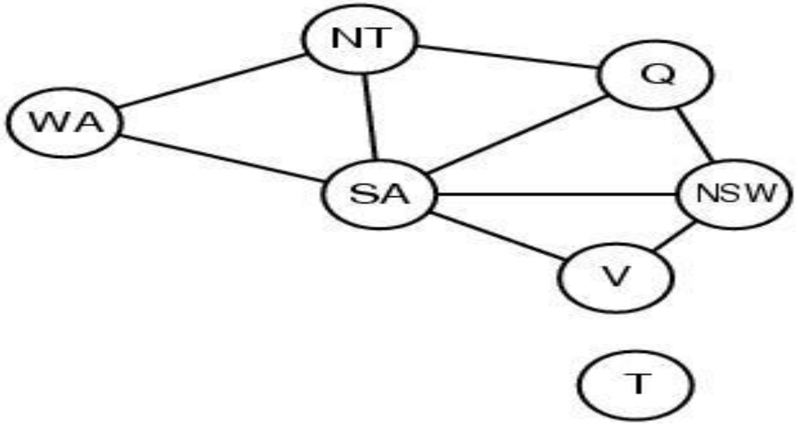
- Use of min-conflicts heuristic in hill-climbing.

Min-conflicts example 2



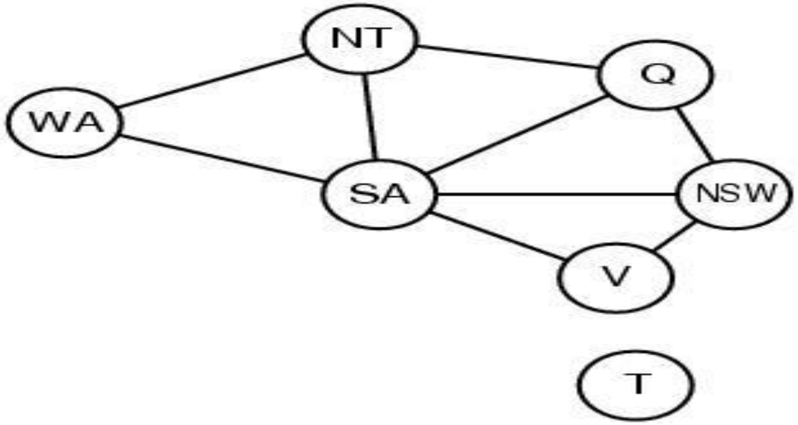
- A two-step solution for an 8-queens problem using min-conflicts heuristic.
- At each stage a queen is chosen for reassignment in its column.
- The algorithm moves the queen to the min-conflict square breaking ties randomly.
- Result

Problem structure



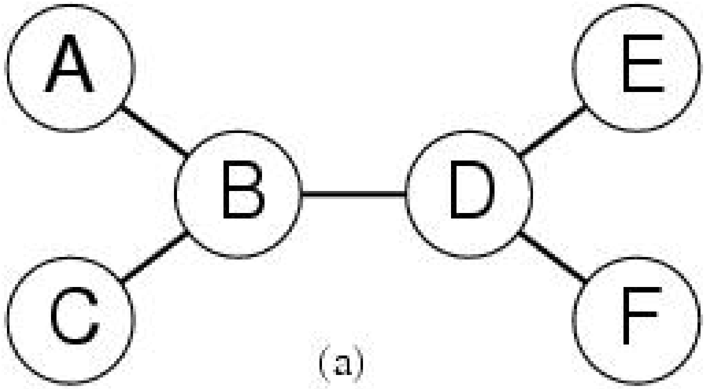
- *How can the problem structure help to find a solution quickly?*
- Subproblem identification is important:
  - **Coloring Tasmania and mainland are independent subproblems**
  - **Identifiable as connected components of constrained graph.**
- Improves performance

Problem structure



- Suppose each problem has *c* variables out of a total of *n*.
- Worst case solution cost is $O(n/c\ d^c)$, i.e. linear in *n*
  - **Instead of $O(d^n)$, exponential in *n***
- E.g. $n= 80,\ c= 20,\ d=2$
  - **2^{80} = 4 billion years at 1 million nodes/sec.**
  - **$4 * 2^{20}$= .4 second at 1 million nodes/sec**

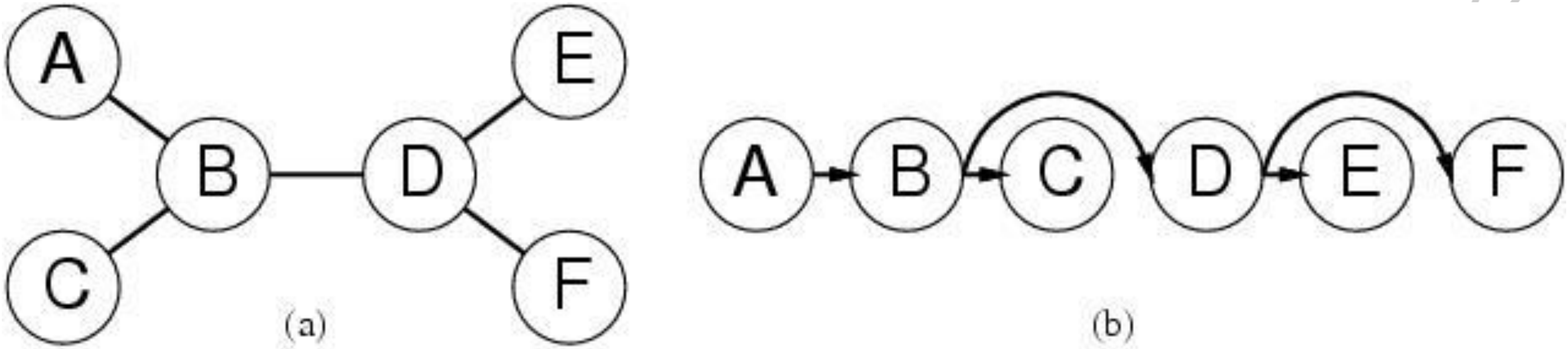
Tree-structured CSPs



(a)

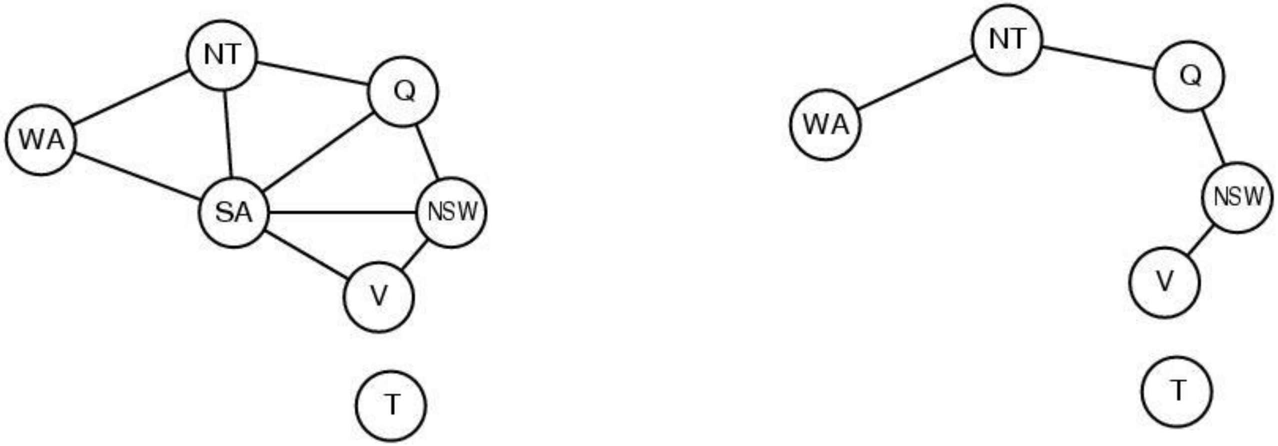
- Theorem: if the constraint graph has no loops then CSP can be solved in $O(nd^2)$ time
- Compare difference with general CSP, where worst case is $O(d^n)$

Tree-structured CSPs



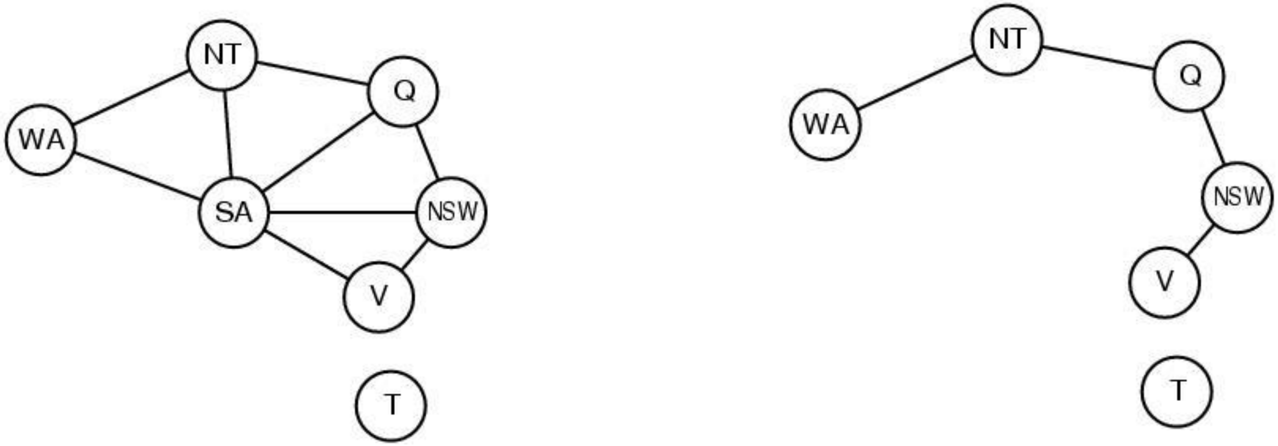
- In most cases subproblems of a CSP are connected as a tree
- Any tree-structured CSP can be solved in time linear in the number of variables.
  - **Choose a variable as root, order variables from root to leaves such that every node's parent precedes it in the ordering.**
  - **For *j* from *n* down to 2, apply REMOVE-INCONSISTENT-VALUES(Parent(X_j),X_j)**
  - **For *j* from 1 to *n* assign X_j consistently with Parent(X_j)**

Nearly tree-structured CSPs



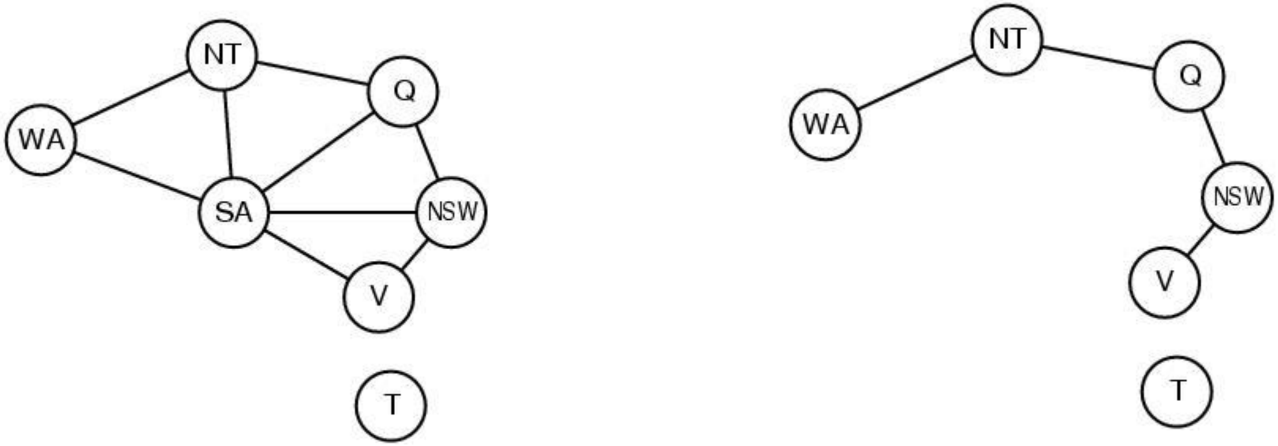
- *Can more general constraint graphs be reduced to trees?*
- Two approaches:
  - **Remove certain nodes**
  - **Collapse certain nodes**

Nearly tree-structured CSPs



- Idea: assign values to some variables so that the remaining variables form a tree.
- Assume that we assign *{SA=x}* $\leftarrow$ *cycle cutset*
    - **And remove any values from the other variables that are inconsistent.**
    - **The selected value for SA could be the wrong one so we have to try all of them**

Nearly tree-structured CSPs



- This approach is worthwhile if cycle cutset is small.
- Finding the smallest cycle cutset is NP-hard
  - **Approximation algorithms exist**
- This approach is called *cutset conditioning*.

Summary

- CSPs are a special kind of problem: states defined by values of a fixed set of variables, goal test defined by constraints on variable values
- Backtracking=depth-first search with one variable assigned per node
- Variable ordering and value selection heuristics help significantly
- Forward checking prevents assignments that lead to failure.
- Constraint propagation does additional work to constrain values and detect inconsistencies.
- The CSP representation allows analysis of problem structure.
- Tree structured CSPs can be solved in linear time.
- Iterative min-conflicts is usually effective in practice.


THE END.